

RAHIM HOSSAIN

Applied Machine Learning Researcher · Multimodal Retrieval & Vision–Language Models

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SUMMARY

Applied ML researcher and Computer Science PhD candidate specializing in multimodal retrieval and vision–language models. Industry experience at eBay Search, building and adapting VLMs (BLIP, CLIP, SigLIP2, Qwen-VL) for large-scale product search on industry GPU clusters. First-author papers under review at NeurIPS and in submission to ICLR; publications at ICCV, IJCNN, and Sensors. Winner of eBay’s 2026 company-wide intern hackathon, recognized by the CEO.

EXPERIENCE

eBay — Applied Research Intern, Search Team

San Jose, CA · May 2026 – Aug 2026

- Fine-tuned a **BLIP** vision–language model with **visual-grounding late interaction** on eBay Fashion catalog data, delivering significant retrieval gains over the existing production retrieval model .
- Ran large-scale training and evaluation on **eBay’s industry GPU clusters** over industry-scale labeled product data building the data and evaluation pipelines end to end.
- Designed **Tip-Region**, a gradient-free few-shot adaptation method that adds a paired text–region exemplar cache to frozen backbones; improved localized retrieval across CLIP, SigLIP2, and Qwen3-VL-Embedding at every recall level — **first-author paper in submission to ICLR 2027**.
- Uncovered and corrected silent train–test contamination in a public benchmark mirror by building a **support-set leakage audit**, and built LocFG, a leakage-controlled benchmark for small-referent retrieval.
- **Won the eBay Vibe Coding Hackathon** (champion among all intern teams company-wide); recognized by **CEO Jamie Iannone**, with a follow-up engagement from eBay’s marketing team on the winning project.

SUNY Binghamton — Graduate Research Assistant, RT Lab

Binghamton, NY · May 2023 – Present

- Developed **MM++**, an unsupervised, scale-invariant multilayer OOD detection method via top-K gated feature fusion — **first-author paper under review at NeurIPS 2026**.
- Built an energy-efficient deep learning pipeline for spotted-lanternfly detection over LoRa wide-area networks.

Cobalt Speech & Language, Inc. — Research Scientist

Remote · Mar 2022 – Dec 2022

- Core contributor to the Call Summarization product: fine-tuned and evaluated transformer models (BART, BERT, LED) for long-context summarization and natural-language understanding.

EDUCATION

SUNY Binghamton — PhD, Computer Science

Aug 2024 – Present

SUNY Binghamton — MS, Computer Science

Jan 2023 – May 2024

Bangladesh University of Engineering & Technology (BUET) — BSc, EEE

Apr 2016 – Jan 2022

PUBLICATIONS

- First-author manuscript on gradient-free few-shot adaptation for localized vision–language retrieval — **in submission to ICLR 2027** (eBay internship research).
- R. Hossain et al. “MM++: Unsupervised Scale-Invariant Multilayer OOD Detection via Top-K Gated Feature Fusion.” **Under review at NeurIPS 2026**. [arXiv:2606.17352](https://arxiv.org/abs/2606.17352).
- S. Ahmed, A. A. Arafat, M. N. Rizve, R. Hossain, Z. Guo, A. S. Rakin. “SSDA: Secure Source-Free Domain Adaptation.” *ICCV*, 2023.
- M. D. Samad, R. Hossain, K. M. Iftekharuddin. “Dynamic Perturbation of Weights for Improved Data Reconstruction in Unsupervised Learning.” *IJCNN*, 2021.

SKILLS

Languages: Python, C/C++, Go, MATLAB

ML frameworks: PyTorch, Hugging Face Transformers, TensorFlow, Keras, Scikit-Learn

ML domains: Multimodal retrieval, vision–language models (CLIP, SigLIP, BLIP, Qwen-VL), late interaction, few-shot & test-time adaptation, NLP / LLMs

Infrastructure: Distributed GPU-cluster training, large-scale data & evaluation pipelines, Linux

AWARDS & HONORS

- **eBay Vibe Coding Hackathon — Champion (2026):** champion among all intern teams; recognized by eBay’s CEO.
- eBay ML Challenge — 2nd place (2025) · 3rd place (2024).
- Industrial Automation Challenge (national robotics competition, Bangladesh) — 2nd place.